



ANGELES UNIVERSITY FOUNDATION

Mac Arthur Highway, Angeles City, Philippines, 2009

BSIT 3A - A.Y. 2026-2027

Lab IV: Decorator Reflections

For this lab, I learned how Decorator pattern can add new features to an object without making many subclasses. In the bad version, there are different classes for logging, caching, and other features. If more features is added, the number of subclasses can become a lot because the features need to be combine.

In the fixed version, I used an HttpClient interface and an HttpDecorator that can wrap another HttpClient. I made different decorators for Logging, Caching, Retry, and Token Counter. Each decorator have its own job and they can be combined together.

I also learned that the order of the decorators can affect the result. For example, logging and caching can behave differently depending on which one is placed first. This was useful for me because I saw that decorators are not only about adding features, but also how the features are arranged.

According to Gamma et al. (1994), Decorator can attach additional responsibilities to an object dynamically. In this lab, I was able to add different behaviors by wrapping the objects instead of creating a lot of subclasses. This makes the code more easier to add new features.

